The exact maximum in Erdős problem 272 for sufficiently large ground sets

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Abstract

Let $v(N)$ be the maximum cardinality of a family of distinct subsets of $\{1,\dots,N\}$ whose intersections of distinct members are nonempty arithmetic progressions. We give a computer-assisted proof that

$$v(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor \qquad (N \geq 2^{49}).$$

The lower bound is Szabó's construction. For the upper bound we delete large non-progressions, obtain a common point for the remaining nonprogressions, and force all members avoiding that point to be long intervals. Their common interval then pays for the deleted members. We give the explicit numerical estimates and the precise interfaces of the formally proved counting lemmas. The accompanying Lean development is part of the proof. More refined computations in that development establish the same formula for every $N \geq 642498$; those refinements are not developed in this paper. The remaining finite range is not settled here.

1 Statement and scope

Write $[N] = \{1,\dots,N\}$. An arithmetic progression (AP) means a nonempty finite set $\{a, a+d, \dots, a+kd\}$ with $d \geq 1$ and $k \geq 0$. Thus singletons and pairs are APs. A family $\mathcal{F} \subseteq 2^{[N]}$ is *legal* if $S \cap T$ is an AP whenever $S, T \in \mathcal{F}$ and $S \neq T$. The members themselves need not be APs. Set

$$B(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor, \qquad v(N) = \max_{\mathcal{F} \text{ legal}} |\mathcal{F}|.$$

The maximum exists because its domain is finite.

Theorem 1. *For every integer $N \geq 2^{49}$, $v(N) = B(N)$. For every $N \geq 1$, there is a legal family of size $B(N)$.*

This is a partial answer to Erdős problem 272 and the corresponding Justin Sun Prize problem JSP-000233. Szabó established the leading term and the construction giving $B(N)$ [1]. Yang formulated the exact formula and proved the common-point bound using private pairs and a staircase inequality [2]. Our formal development also follows the non-common-point structural route of Kohlmeyer and Kruer [3]. We do not claim priority for those constructions or methods. We prove existence of an optimal common-point family, not that every optimal family has a common point.

*Project and complete proof sources: https://github.com/lixiang90/Erdos272.

The theorem is *computer-assisted*: the finite-set counting lemmas specified below have complete proofs in the companion Lean source, which forms an electronic supplement to this paper. The paper supplies their mathematical interfaces and the full numerical and logical assembly. It does not replace those source proofs by an assertion that a numerical search succeeded. Section 5 gives direct paths and reproduction commands.

2 The matching construction

Let $c = \lceil N/2 \rceil$ and $k = \lfloor (N-1)/4 \rfloor$. Start with every subset of $[N]$ of size at most three containing c. For each $1 \le d \le k$, remove the two triples

$$\{c-2d, c, c+d\}, \qquad \{c-d, c, c+2d\},$$

and add the five-term progression $P_d = \{c-2d, c-d, c, c+d, c+2d\}$ and its two four-term end deletions.

Proposition 2 (Szabó's construction)**.** *For every integer $N \ge 1$, the resulting family is legal and has cardinality $B(N)$.*

Proof. Since $2k \le \min(c-1, N-c)$, all its points lie in $[N]$. Every member contains c. Intersections involving a member of size at most two, or two distinct triples, therefore have size one or two. Two APs have an AP as their nonempty intersection: their common points form a residue class modulo the least common multiple of their steps, restricted to their common span.

It remains to intersect a retained triple with an added AP. Only a three-point intersection needs checking. Such a triple lies in some P_d and contains c. After translating and scaling, the six possible triples through zero in $\{-2,-1,0,1,2\}$ consist of four APs and the two deleted triples $\{-2,0,1\}$ and $\{-1,0,2\}$. Thus every required intersection is an AP.

The ordered gaps $(2d, d)$ and $(d, 2d)$ distinguish the deleted triples. The step and endpoints distinguish the added APs. Consequently exactly $2k$ triples are removed and $3k$ larger members are added, giving

$$N + \binom{N-1}{2} - 2k + 3k = \binom{N}{2} + 1 + k.$$

This also covers $k = 0$. □

3 The structural counting certificates

A member is *crooked* if it is not an AP. For a legal family $\mathcal{F}$, define

$$\mathcal{D}(\mathcal{F}) = \{S \in \mathcal{F} : S \text{ is crooked and } 28|S| > N\}, \qquad \mathcal{G}(\mathcal{F}) = \mathcal{F} \setminus \mathcal{D}(\mathcal{F}).$$

Every AP member is retained. Write $H_N = \sum_{i=1}^{N} 1/i$ and $J = \lfloor \log_2 N \rfloor + 1$. All intervals below are integer intervals, with both endpoints included.

We use the following formally proved structural facts. Their hypotheses refer to the actual family and its subsets. In particular they do not assume an upper bound on an abstract collection of witnesses.

Lemma 3 (Preparation and endpoint deficit)**.** *Suppose $N \ge 2^{49}$, $\mathcal{F}$ is legal, and $|\mathcal{F}| > B(N)$. Either leave $\mathcal{F}$ unchanged or reflect every member by $x \mapsto N+1-x$, obtaining a legal family $\mathcal{H}$ with $|\mathcal{H}| = |\mathcal{F}|$. There are integers $1 \le c < L \le R \le N$ with the following properties. Put $\mathcal{D} = \mathcal{D}(\mathcal{H})$, $\mathcal{G} = \mathcal{G}(\mathcal{H})$, and $\mathcal{A} = \{S \in \mathcal{G} : c \notin S\}$.*

1. *Every crooked member of $\mathcal{G}$ contains c, and $\mathcal{A} \neq \varnothing$.*

2. *Every $S \in \mathcal{A}$ has $\min S \leq L$ and $\max S \geq R$.*

3. *With $t = L + N - R$, there is a nonnegative witness count E such that*

$$|\mathcal{D}| \leq 3137N, \qquad E \leq 6NH_N^3, \tag{1}$$

$$|\mathcal{H}| \leq \frac{N^2}{2} - \frac{t^2}{102} + |\mathcal{D}| + E + 40NH_N^2 + 6NH_N + 24N + (L - c) + 3. \tag{2}$$

Here and below the statements of the source lemmas are stronger than the specializations displayed. For Lemma 3, the exact source is `Counting/MiddleDeficit52.lean`, declaration `actual_original_prepared_endpoint102`; deletion and the exceptional count are supplied by `Core/DenseSharper41.lean` and `Arithmetic/DivisorMoment45.lean`. All source paths in the paper are relative to `formal/Erdos272/`.

We describe the counting mechanism to clarify what is certified. A crooked set containing a root c has a pair $\{x, y\}$ for which the smallest AP containing c, x, y is not contained in the set. Such a pair cannot belong to another member through c, since their legal intersection would contain that AP. These private pairs, and the pairs covered by AP members, are counted in the same pair universe. Deleting large crooked members first bounds their cost by $(4 \cdot 28^2 + 1)N = 3137N$. The remaining crooked members have a common point by the empty-kernel counting bound. If the retained family itself has a common point, the common-point bound and the omitter estimate already exclude a counterexample. Otherwise an avoiding AP supplies a one-sided orientation, and endpoint extrema supply L, R.

The endpoint estimate counts primitive pairs in the middle region; coprimality, floor errors, and exceptional witnesses are retained in (2). The coefficient $1/102$ is deliberately conservative. These combinatorial arguments, including the common-point bound and its finite staircase certificate and unbounded tail, are proved in the import closure of the cited module. No kernel conjecture is assumed.

Lemma 4 (Excluding noninterval avoiders)**.** *Let $\mathcal{G}$ be legal, let $c \in [N]$, and suppose every crooked member of $\mathcal{G}$ contains c. Suppose $c \leq L \leq N$, and each member avoiding c contains a point at most L. If an avoiding member is not an integer interval, then*

$$|\mathcal{G}| \leq \frac{3N^2}{8} + \frac{5}{3}LN + 2NH_N + 20N + 3 + 6NH_N^3. \tag{3}$$

In particular, if $N \geq 6$, $N < 2^J$, $J \geq 2$, $32L \leq N$, and $16384J^3 \leq N$, then $|\mathcal{G}| \leq 11N^2/24$.

The exact finite-set counting proof is `Global/NonInterval53.lean`, declarations `actual_noninterval_family_count` and `actual_noninterval_family_gap`. An avoiding member is an AP. If it is not an interval, its step is at least two and its size is at most $N/2 + 1$. The unordered pairs meeting it number at most $3N^2/8 + N$. Private witnesses for crooked members inject into these pairs, except for the separately bounded exceptional members. Counting the AP members with a low endpoint gives (3). For the stated consequence, use $H_N \leq J$ and $J \geq 2$. The last four terms in (3) are at most $31NJ^3 \leq 31N^2/16384$. Hence

$$\frac{3}{8} + \frac{5}{96} + \frac{31}{16384} < \frac{11}{24}.$$

The elementary inequality $H_N \leq J$ follows by partitioning $[N]$ into the dyadic blocks $\{1\}, [2, 3], [4, 7], \ldots$; each contributes at most one.

Lemma 5 (Interval count and restoration)**.** *Let $N \ge 29^3$ and $1 \le c < L \le R \le N$, and put $t = L + N - R$. Let $\mathcal{H}$ be legal, $\mathcal{D} = \mathcal{D}(\mathcal{H})$, and $\mathcal{G} = \mathcal{G}(\mathcal{H})$. Suppose the members of $\mathcal{G}$ avoiding c form a nonempty family of intervals $[a, b]$ satisfying $c < a \le L \le R \le b \le N$. If*

$$6tJ \le N, \qquad 812t \le N, \tag{4}$$

then

$$|\mathcal{G}| \le \binom{N}{2} + 1, \qquad |\mathcal{D}| \le 131(t-1) \le \left\lfloor \frac{N-1}{4} \right\rfloor. \tag{5}$$

Consequently $|\mathcal{H}| \le B(N)$.

The first bound is `Retention/Terminal36.lean`, declaration `actual_native_interval_terminal`. The second is `Retention/JointFringe44.lean`, declaration `actual_dense_fringe_bound`, with parameters $r = 28$, $p = 29$. Their assembly is `Global/GlobalUpper53.lean`, declaration `actual_deleted_long_core_restore28`.

Here is why the two counts fit together. An avoiding interval contains $[L, R]$, so this core lies in an actual AP member of the original family. Its exterior in $[N]$ has cardinality $t - 1$. A deleted member has more than $N/28$ points. The inequality $28 \cdot 29t \le N$ forces a long trace in the core. Bounded-step trace anchors and points outside the core give the joint bound

$$\mathrm{JointCount}(29)(t-1) = \left(29 + \sum_{d=1}^{28} \left\lfloor \frac{29}{d} \right\rfloor\right)(t-1) = 131(t-1).$$

The aligned and off-residue cases are both included in this count. For the final payment, $4 \cdot 131t = 524t \le 812t \le N$ implies $4 \cdot 131(t-1) \le N - 1$. Since $\mathcal{D}$ and $\mathcal{G}$ partition the original family, their bounds add to exactly $B(N)$.

4 The explicit large-order argument

We now prove Theorem 1 from the certified counting lemmas. Every numerical estimate in this section is elementary.

Lemma 6 (Uniform logarithmic payments)**.** *If $N \ge 2^{49}$ and $J = \lfloor \log_2 N \rfloor + 1$, then*

$$J \ge 50, \qquad 470771616J^3 \le N, \qquad 25704J^5 \le N. \tag{6}$$

Proof. Put $k = J - 1 \ge 49$. At $k = 49$, direct integer arithmetic gives

$$470771616 \cdot 50^3 = 58846452000000 < 2^{49}, \qquad 25704 \cdot 50^5 = 8032500000000 < 2^{49}.$$

For $k \ge 49$, $(k+2)/(k+1) \le 51/50$, and $(51/50)^5 < 2$. Thus both $(k+1)^3/2^k$ and $(k+1)^5/2^k$ decrease by induction. The desired bounds follow from $2^k \le N$. □

Proof of Theorem 1. Proposition 2 proves the lower bound. Suppose for a contradiction that $N \ge 2^{49}$ and $|\mathcal{F}| > B(N)$ for a legal family. Apply Lemma 3 and use its notation. Since $H_N \le J$, $L - c \le N$, and $N \ge 1$, (1)–(2), together with $|\mathcal{H}| \ge \binom{N}{2}$, imply

$$t^2 \le 102N(6J^3 + 40J^2 + 6J + 3166) \le 714NJ^3. \tag{7}$$

For the last inequality, $J \ge 50$ and the identity

$$J^3 - 40J^2 - 6J - 3166 = (J - 50)(J^2 + 10J + 494) + 21534 \ge 0$$

suffice. The slack in 3166 also covers the $N/2$ difference between $N^2/2$ and $\binom{N}{2}$, and the constant 3 in (2).

As $812^2 \cdot 714 = 470771616$ and $36 \cdot 714 = 25704$, Lemma 6 and (7) give

$$(812t)^2 \le N^2, \qquad (6tJ)^2 \le N^2.$$

All quantities are nonnegative, so $812t \le N$ and $6tJ \le N$. Also $L \le t$, whence $32L \le N$, and $16384J^3 \le 470771616J^3 \le N$.

If an avoiding member of $\mathcal{G}$ were not an interval, Lemma 4 would give $|\mathcal{G}| \le 11N^2/24$. On the other hand, $|\mathcal{H}| > B(N)$ and $|\mathcal{D}| \le 3137N$ imply

$$\frac{N^2}{2} \le |\mathcal{G}| + 3138N.$$

These inequalities would force $N \le 24 \cdot 3138 = 75312$, contrary to $N \ge 2^{49}$. Therefore every avoiding member is an integer interval $[a, b]$. The endpoint conditions in Lemma 3 give $a \le L \le R \le b$. Since $c < L$ and $c \notin [a, b]$, necessarily $c < a$.

All hypotheses of Lemma 5 now hold, for the same family, root, and endpoints. It follows that $|\mathcal{F}| = |\mathcal{H}| \le B(N)$, a contradiction. □

5 Formal verification and the stronger range

The repository is https://github.com/lixiang90/Erdos272. The public entry is `formal/PublicResults.lean`. It defines the ground set, AP predicate, legality, $B(N)$, the finite set of legal families, and its extremal value directly in that file using Lean and mathlib types. The proofs apply to these definitions by definitional equality with the internal finite-set model. In particular there is no additional common-point hypothesis in the public upper bound.

Corollary 7 (Refined formal result)**.** *For every integer $N \ge 642498$, $v(N) = B(N)$, and a legal family attaining this value exists.*

This stronger result uses more refined arithmetic and finite certificates than the argument developed above. It is not proved in detail in this paper; readers can inspect and check the complete source through `PublicResults.exact_value` and `PublicResults.optimal_family` in namespace `JSP233`. The paper's own theorem has the separate entry `PublicResults.paper_exact_value`, which calls the 2^{49} argument directly. Thus the simpler theorem can be traced without relying on the refined threshold.

The development is pinned to Lean 4.35.0 and mathlib's matching stable release. The lock file records the exact dependency revisions. From a checkout with the specified Lean toolchain, run

```
cd formal
lake exe cache get
lake build
```

The public entry prints the theorem types and their axiom dependencies. The proofs use only `propext`, `Classical.choice`, and `Quot.sound`; finite certificates are checked by Lean's kernel. A full build can require substantial time and memory. With limited resources, compile the project modules sequentially in dependency order, reuse compiled dependencies, and choose process, thread, memory, and CPU limits suited to the available hardware.

The directory groups separate foundational set geometry, private-pair counting, retention and restoration, arithmetic certificates, and global assembly. A proof covering the entire remaining finite range is still missing.

Acknowledgements and collaboration. The project builds on the work cited below and was developed with AI-assisted research, programming, and review. Formal checking and independent review serve different purposes: the former checks the proof terms, and the latter checks their interpretation as the stated mathematical problem. Contributions that close the remaining finite gap are welcome. Please credit the earlier results and all new contributions, and coordinate a joint application for the Justin Sun Prize.

References

[1] T. Szabó, Intersection properties of subsets of integers, *European Journal of Combinatorics* **20** (1999), 429–444. https://page.mi.fu-berlin.de/szabo/PDF/aps.pdf.

[2] Z. Yang, Exact values and exact upper bounds for families of integers with arithmetic progression intersections (Erdős Problem #272), arXiv:2607.23004v1 (2026). https://arxiv.org/abs/2607.23004v1.

[3] J. Kohlmeyer and L. Kruer, *A linear error bound for Erdős problem 272*, manuscript and companion Lean development, fixed source commit `a624e9de77e0bd8868e7103f8b379f9dbd86b607` in JENW1N/complete-math-solutions (fixed Lean source). The source informed the structural route; the present theorems are checked in the accompanying finite-integer-set model.

[4] The Justin Sun Prize, Awards and problem list, https://github.com/TheJustinSunPrize/awards.